



2024 BSC CS...



FACULTY OF SCIENCE AND HUMANITIES

ACADEMIC CURRICULA

UNDERGRADUATE DEGREE PROGRAMME

Bachelor of Science (Honors)

in

Computer Science

Four Years

National Education Policy

Learning Outcome-based Curricula Framework

National Credit Framework

Academic Year

2024 - 2025



SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

(Deemed to be University u/s 3 of UGC Act, 1956)

Kattankulathur, Chengalpattu District 603203, Tamil Nadu, India

FACULTY OF SCIENCE AND HUMANITIES
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

1.	Vision Statement
Stmnt - 1	To reach the goals of expansion and improvement of the department and rank among the top computer science departments in the world for teaching, learning, research, and community involvement
Stmnt - 2	Ensure that students have high-quality learning experiences by implementing efficient classroom procedures, using active learning techniques, and creating chances for students and faculty to engage in meaningful interactions



7. Programme Structure

Discipline Core Courses (C)				
Course Code	Course Title	Hours/Week		
		L	T	P
				C
UCS24101J	Digital Electronics	3	0	3
USA24102J	Programming for Problem Solving	3	0	3
UMS24101T	Discrete Mathematical Structures	4	0	0
USA24201J	Data Structures and Algorithms	3	0	3
UCS24202J	Object Oriented Programming using Java	3	0	3
UMS24202T	Mathematical Foundation	4	0	0
UCS24301J	Operating System	3	0	3
USA24302J	Database Management System	3	0	3
UMS24301T	Statistical Methods	4	0	0
USA24401J	Python Programming	3	0	3
Total Learning Credits				40

Elective Courses (Minor) (E)				
Course Code	Course Title	Hours/Week		
		L	T	P
				C
UDS24B01J	Data Science and Analytics	3	0	3
UDS24B02J	Data Analytics using Spreadsheet	3	0	3
UDS24B03J	Essentials of Machine Learning	3	0	3
UDS24B04J	Programming using R	3	0	3
UDS24B05J	Programming using Python	3	0	3
UDS24B06J	Introduction to Deep Learning	3	0	3
UDS24B07J	Applications of Computer Vision	3	0	3
UDS24B08J	Data Visualization Tools	3	0	3
Total Learning Credits				32

Ability Enhancement Courses (A)				
Course Code	Course Title	Hours/Week		
		L	T	P
				C
ULT24AE1J	Tamil - I	2	0	2
ULT24AE1J	Hindi - I	2	0	2
ULT24AE1J	French - I	2	0	2
ULT24AE1J	English	2	0	2
ULT24AE2J	Tamil - II	2	0	2
ULT24AE2J	Hindi - II	2	0	2
ULT24AE2J	French - II	2	0	2
Total Learning Credits				9

Discipline Core/ Elective Courses (D)				
Course Code	Course Title	Hours/Week		
		L	T	P
				C
UMS24D01T	Resource Management Techniques	4	0	0
UMS24D02T	Numerical Methods	4	0	0
UCS24D01J	Software Engineering	3	0	3
UCS24D02J	Computer Networks and Data Communication	3	0	3
UCS24D03J	Object Oriented Analysis and Design	3	0	3
UCS24D04J	Machine Learning	3	0	3
UCS24D05J	Fundamentals of Business Application Development	3	0	3
UCS24D06J	NLP and Language Models	3	0	3
USA24D07J	Cloud Computing	3	0	3
UCS24D08J	Microcontrollers and Embedded Systems	3	0	3
UCS24D09J	Mobile Application Development	3	0	3
USA24D10T	Wireless Communication and Mobile Computing	4	0	0
UCS24D11J	Web Application Framework	3	0	2
UCS24D12J	Computer Vision	3	0	2
UCS24D13J	Big Data Analytics	3	0	2
UCS24D14J	Cryptography and Network Security	3	0	2
UCS24D15J	AI and Business Analytics	3	0	2
USA24D16J	Cyber Security	3	0	2
USA24D20J	Blockchain Technology	3	0	2
UCS24D18T	Introduction to Cyber Physical Systems	4	0	0
Total Learning Credits				40

Multidisciplinary Courses (M)				
Course Code	Course Title	Hours/Week		
		L	T	P
				C
UCS24M01J	Introduction to ICT Tools	2	0	2
UCS24M02J	Fundamentals of Digital Marketing	3	0	1
UCS24M03J	Basics of Web Application Development	2	0	2
Total Learning Credits				9

Value Added Courses (V)				
Course Code	Course Title	Hours/Week		
		L	T	P
				C
UCD24V01T	Essentials of Artificial Intelligence	1	0	0
UCS24V01T	Environmental Studies	2	0	0
UCD24V02T	Universal Human Values	2	0	0
Total Learning Credits				5

Skill Enhancement Courses (S)				
Course Code	Course Title	Hours/Week		
		L	T	P
				C
UCD24S01J	Verbal Ability and Skill Development	1	0	2
UCD24S02J	Quantitative Aptitude and Logical Reasoning	0	0	2
UCD24S03J	Industry Oriented Employability and Leadership Skills	1	0	2
UCD24S04J	Communication Skills	0	0	2
UCD24S05J	Career Readiness and Professional Skills	1	0	2
Total Learning Credits				9



Project Work / Internship (P)				
Course Code	Course Title	Hours/Week		
		L	T	P
				C
UCS24P01J	Internship - I	-	-	-
UCS24P02J	Internship - II	-	-	-
UCS24P03J	Project Work	0	0	4
UCS24P04J	Research Project and Dissertation	0	0	24
UCS24P05J	Professional Internship	0	0	4
Total Learning Credits				18

Mandatory Courses (Y)				
Course Code	Course Title	Hours/Week		
		L	T	P
				C
UNIS24Y01L	NS	0	0	0
UNIS24Y02L	NCC	0	0	0
UNIS24Y03L	NSO	0	0	0
UNIS24Y04L	YOGA	0	0	0
UNIS24Y05L	My India Project	0	0	0
Total Learning Credits				0

8. Course Allocation across Semesters

Semester – I					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
ULT24AE1J	Tamil – I	2	0	2	3
ULH24AE1J	Hindi – I				
ULF24AE1J	French – I				
ULE24AE1J	English	2	0	2	3
UCS24101J	Digital Electronics	3	0	3	4
USA24102J	Programming for Problem Solving	3	0	3	4
UMS24101T	Discrete Mathematical Structures	4	0	0	4
UCD24S01J	Verbal Ability and Skill Development	1	0	2	2
Learning Credits					20

Semester – II					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
ULT24AE2J	Tamil – II	2	0	2	3
ULH24AE2J	Hindi – II				
ULF24AE2J	French – II				
USA24201J	Data Structures and Algorithms	3	0	3	4
UCS24202J	Object Oriented Programming using Java	3	0	3	4
UMS24202T	Mathematical Foundation	4	0	0	4
UCD24V01T	Essentials of Artificial Intelligence	1	0	0	1
	Multidisciplinary Course – I				3
UCD24S02L	Quantitative Aptitude and Logical Reasoning	0	0	2	1
UNS24Y01L	NSS	0	0	0	0
UNC24Y01L	NCC				
UNO24Y01L	NSO				
UYG24Y01L	YOGA				
Learning Credits					20

Total Learning Credits of Courses of FIRST Year = 40

Bachelor's Programme, Regulation 2024

6

Semester – III					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UCS24301J	Operating System	3	0	3	4
USA24302J	Database Management System	3	0	3	4
UMS24303T	Statistical Methods	4	0	0	4
	Minor Elective – I				4
UCS24P01L	Internship – I	0	0	0	1
	Multidisciplinary Course – II				3
Total Learning Credits					20

Semester – III					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UCS24301J	Operating System	3	0	3	4
USA24302J	Database Management System	3	0	3	4
UMS24303T	Statistical Methods	4	0	0	4
	Minor Elective – I				4
UCS24P01L	Internship – I	0	0	0	1
	Multidisciplinary Course – II				3
Total Learning Credits					20

Semester – IV					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
USA24401J	Python Programming	3	0	3	4
UMS24D01T	Resource Management Techniques	4	0	0	4
UMS24D02T	Numerical Methods				4
	Minor Elective – II				4
	Minor Elective – III				4
UEN24S01L	Communication Skills	0	0	4	2
UCD24S03J	Industry Oriented Employability and Leadership Skills	1	0	2	2
UMI24Y01L	My India Project	0	0	0	0
Total Learning Credits					20

Total Learning Credits of Courses of SECOND Year = 40

Bachelor's Programme, Regulation 2024

7



FACULTY OF SCIENCE AND HUMANITIES
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY



Semester – V					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UCS24D01J	Software Engineering	3	0	3	4
UCS24D02J	Computer Networks and Data Communication				
UCS24D03J	Object Oriented Analysis and Design				
UCS24D04J	Machine Learning	3	0	3	4
	Minor Elective – IV				4
	Multidisciplinary Course – III				3
UES24V01T	Environmental Studies	2	0	0	2
UCD24S04J	Career Readiness and Professional Skills	1	0	2	2
UCS24P02L	Internship – II	0	0	0	1
Total Learning Credits					20

Semester – VI					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UCS24D05J	Fundamentals of Business Application Development	3	0	3	4
UCS24D06J	NLP and Language Models				
USA24D07J	Cloud Computing	3	0	3	4
UCS24D08J	Microprocessors and Embedded Systems				
	Minor Elective – V				4
	Minor Elective – VI				4
UCS24P03L	Project Work	0	0	4	2
UCD24V02T	Universal Human Values	2	0	0	2
Total Learning Credits					20

Total Learning Credits of Courses of THIRD Year = 40

Semester – VII					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UCS24D09J	Mobile Application Development	3	0	3	4
USA24D10T	Wireless Communication and Mobile Computing	4	0	0	
UCS24D11J	Web Application Framework	3	0	2	
UCS24D12J	Computer Vision	3	0	2	
UCS24D13J	Big Data Analytics	3	0	2	4
UCS24D14J	Cryptography and Network Security				
	Minor Elective – VII				4
	Minor Elective – VIII				4
Learning Credits					20

Semester – VIII					
Course Code	Course Title	Hours/ Week			C
		L	T	P	
UCS24D15J	AI and Business Analytics	3	0	2	4
USA24D18J	Cyber Security	3	0	2	
USA24D20J	Blockchain Technology	3	0	2	4
UCS24D18T	Introduction to Cyber Physical Systems	4	0	0	
UCS24P04L	Research Project and Dissertation	0	0	24	12
UCS24P05L	Professional Internship				
Learning Credits					20

Total Learning Credits of Courses of FOURTH Year = 40

Bachelor's Programme, Regulation 2024

9

8.1 Elective Courses (MINOR) offered by the Department of Computer Science to other Departments (Excluding Department of Computer Science)

Elective Courses (Minor) (E)						
Semester	Course Code	Course Title	Sessions Per Week			C
			L	T	P	
III	UDS24E01J	Data Science and Analytics	3	0	3	4
IV	UDS24E02J	Data Analytics using Spreadsheet	3	0	3	4
IV	UDS24E03J	Essentials of Machine Learning	3	0	3	4
V	UDS24E04J	Programming using R	3	0	3	4
VI	UDS24E05J	Programming using Python	3	0	3	4
VI	UDS24E06J	Introduction to Deep Learning	3	0	3	4
VII	UDS24E07J	Applications of Computer Vision	3	0	3	4
VII	UDS24E08J	Data Visualization Tools	3	0	3	4
Total Learning Credits						32

8.2 Multidisciplinary Courses offered by the Department of Computer Science to all departments (Including Department of Computer Science)

Multidisciplinary Courses (M)						
Semester	Course Code	Course Title	Sessions Per Week			C
			L	T	P	
II	UCS24M01J	Introduction to ICT Tools	2	0	2	3
III	UCS24M02J	Fundamentals of Digital Marketing	3	0	1	3
V	UCS24M03J	Basics of Web Application Development	2	0	2	3
Total Learning Credits						9